

Countersunk Hole

Related terms:

[Stress Concentration](#), [Fasteners](#), [Tensiles](#), [Water Jet Machining](#), [Shape Memory Alloy](#), [Exfoliation](#), [Fiber Damage](#), [Front Panel](#), [Matrix Damage](#)

Nuts, bolts, screws, and washers

Colin H. Simmons, ... Neil Phelps, in [Manual of Engineering Drawing \(Fifth Edition\)](#), 2020

ISO metric hexagon socket countersunk head screws (dimensions in Table 18.5)

The basic requirement for countersunk head screws is that the head should fit into a countersunk hole with as great a degree of flushness as possible. Fig. 18.7 and Table 18.5 give dimensions. To achieve this it is necessary for both the head of the screw and the countersunk hole to be controlled within prescribed limits. The maximum or design size of the head is controlled by a theoretical diameter to a sharp corner and the minimum head angle of 90°. The minimum head size is controlled by a minimum head diameter, the maximum head angle of 92° and a flushness tolerance. The edge of the head may be flat, as shown in Fig. 18.8, or rounded but not sharp edged (see below).

Table 18.5. Dimensions of metric hexagon socket countersunk head screws.

Nominal size D	M3	M4	M5	M6	M8	M10	M12	M16	M20
Head diameter									
A – maximum	6.72	8.96	11.2	13.44	17.92	22.4	26.88	33.6	40.32
B – minimum	5.82	7.78	9.78	11.73	15.73	19.67	23.67	29.67	35.61
Head depth H	1.86	2.48	3.1	3.72	4.96	6.2	7.44	8.8	10.16
Socket size J	2	2.5	3	4	5	6	8	10	12
Key engagement K	1.05	1.49	1.86	2.16	2.85	3.60	4.35	4.89	5.49
Fillet radius									
F – minimum	0.1	0.2	0.2	0.25	0.4	0.4	0.6	0.6	0.8
d – maximum	3.4	4.5	5.5	6.6	9	11	14	18	22

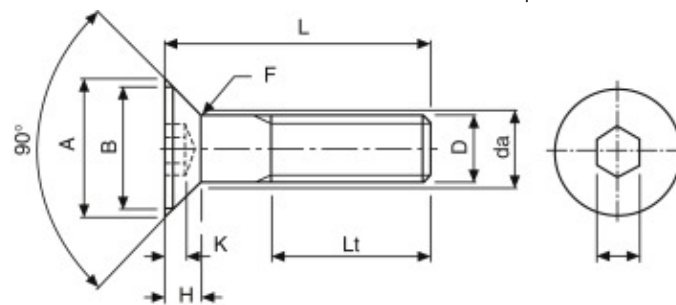


Fig. 18.7. ISO metric hexagon socket countersunk head screws, Dimensions in Table 18.5.

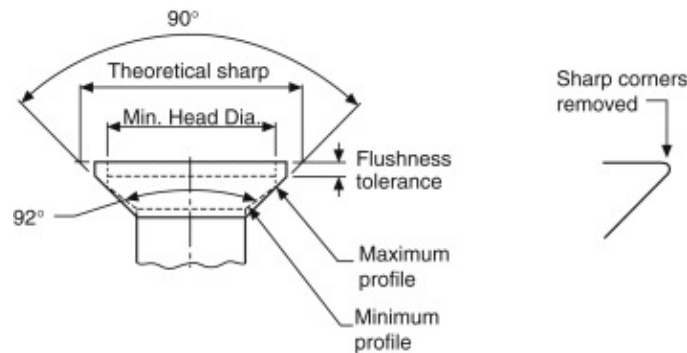


Fig. 18.8. ISO metric hexagon socket countersunk head screws, head details. Dimensions in Table 18.5.

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URL: <https://www.sciencedirect.com/science/article/pii/B9780128184820000189>

Corrosion and fatigue modeling of aircraft structures

N.C. Bellinger, M. Liao, in *Corrosion Control in the Aerospace Industry*, 2009

8.4 Residual fatigue life analysis

Fractographic analyses that were carried out on exfoliated specimens revealed several mechanisms, such as pitting and fretting fatigue, had caused cracks to form, which grew to failure. As mentioned earlier, the HOLSIP framework already contains models to take into account the effect that pitting has on the life of a component. However, owing to a lack of information such as service loading and service environment, these models could not be used to predict the time required to form a pit-like discontinuity in the upper wing skins. Therefore, only a crack growth analysis was carried out to predict the life of a specimen using the 3D finite element (FE) analysis results and the crack growth rate program, AFGROW (Liao *et al.*, 2003a). Since the current version of AFGROW did not contain a crack model at a countersink hole filled with a fastener, the crack model at a straight open hole associated with corrected beta factors was used for this analysis. To estimate these beta factors, 2D FE analyses were carried out to obtain the stress distribution along the assumed crack path for both a straight open hole and the countersunk hole with a fastener. Then the normalized stress values were calculated by dividing the ratio of the stress distribution from the countersunk hole to the straight hole model at the crack origin to the ratios along the assumed crack path. The results indicated that the predicted life was more than 50% less than the test results, which may be attributed to the fact that the nucleation phase was not included.

In another simplified fatigue model, the exfoliation damage was assumed to be a surface crack with a depth that was determined by NDI or a grindout database (Liao *et al.*, 2007). In one simulation, the exfoliation depth was estimated from a grindout database with an assumed lognormal distribution. Using this along with an effective surface crack aspect ratio, a Monte Carlo AFGROW analysis was carried out to calculate the life distributions, the results of which are shown in Fig. 8.9. If